NMAP CHEATSHEET

🛡️ NMAP CHEAT SHEET for Scanning, Enumeration, Vulnerability Finding, and NSE Scripts

🔵 1. Basic Nmap Scans

Command	Purpose
`nmap <target>`	Basic scan
`nmap -sS <target>`	SYN (Stealth) Scan
`nmap -sT <target>`	TCP Connect Scan
`nmap -sU <target>`	UDP Scan
`nmap -p <port> <target>`	Scan specific port
`nmap -p- <target>`	Scan all 65535 ports

🟢 2. Service & Version Detection

Command	Purpose
`nmap -sV <target>`	Detect services and versions
`nmap -sV --version-intensity 9 <target>`	Aggressive version detection

🟡 3. Operating System Detection

Command	Purpose
`nmap -O <target>`	Detect operating system
`nmap -A <target>`	OS detection + version + scripts + traceroute (Aggressive Scan)

🔴 4. Vulnerability Scanning Using Nmap Scripts

Command	Purpose
`nmap --script vuln <target>`	Run default vulnerability detection scripts
`nmap --script http-vuln* -p 80 <target>`	HTTP vulnerability scripts

Command	Purpose
`nmap --script smb-vuln* -p 445 <target>`	SMB related vulnerabilities
`nmap --script ssl-heartbleed -p 443 <target>`	Heartbleed vulnerability scan

🟣 5. NSE (Nmap Scripting Engine) Usage

Step	Command
Go to scripts folder	`cd /usr/share/nmap/scripts/`
List all scripts	`ls` or `ls -l`
Find vulnerability scripts	`ls
Read a specific script	`cat <script-name>.nse` or `less <script-name>.nse`
Update script database	`nmap --script-updatedb`
Run a specific script	`nmap --script <script-name> <target>`

🟤 6. Advanced Scanning Options

Command	Purpose
`nmap -Pn <target>`	Skip host discovery (treat all hosts as up)
`nmap -T4 <target>`	Increase scanning speed (Timing Template 4 - Aggressive)
`nmap -T5 <target>`	Insane speed (may miss some results)
`nmap -f <target>`	Fragment packets (Firewall evasion)
`nmap --top-ports 100 <target>`	Scan top 100 most common ports

🧠 Quick Nmap Concepts

- **-sS:** Stealth scan (Half-open scan)
- **-sV:** Version detection
- **-O:** OS fingerprinting
- **-A:** Aggressive scan (OS + version + scripts + traceroute)
- **-p-:** Scan all ports
- **--script:** Run custom NSE scripts

⚡ HOW TO FIND VULNERABILITIES USING NMAP

1. Direct `vuln` scripts

```
nmap --script vuln <target>
```

2. Manual script selection

```
cd /usr/share/nmap/scripts/
ls | grep vuln
```

Pick specific scripts (e.g., smb-vuln-ms17-010.nse) and run:

```
nmap --script smb-vuln-ms17-010 -p 445 <target>
```

3. Analyze Results

- Check for vulnerabilities, versions, services, open ports.
- Take note of CVEs if scripts mention them.

🧩 ENUMERATION via NMAP

- Find open ports and services:
  ```
  nmap -sV -sC <target>
  ```
- Find SMB shares:
  ```
  nmap --script smb-enum-shares -p 445 <target>
  ```
- Find users via SMB:
  ```
  nmap --script smb-enum-users -p 445 <target>
  ```
- HTTP file discovery:
  ```
  nmap --script http-enum -p 80 <target>
  ```

🚀 MASTER SHORTCUT

```
nmap -A -T4 -p- --script vuln <target>
```

(One-liner aggressive full port + vulnerability scan!)

🔥 EXTRA TIPS

- Always verify with manual enumeration after scanning.
- Nmap can automate basic recon very fast, but never blindly trust — validate manually!

📜 Final Note

> **Nmap is not just a scanner — it's a full information gathering and enumeration powerhouse. Learn it deeply, and you can dominate recon phase like a pro hacker.**